

FARM ANIMALS

Java Programming Assignment

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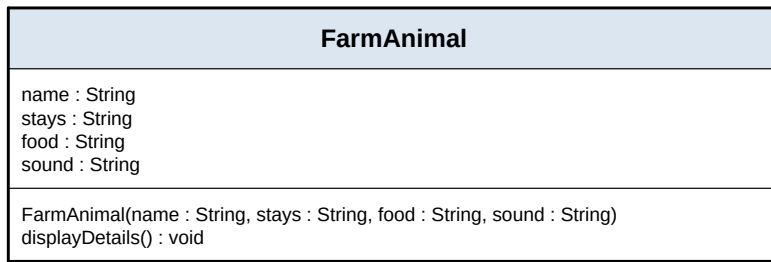
1. ALGORITHM

Steps for executing the FarmAnimals program:

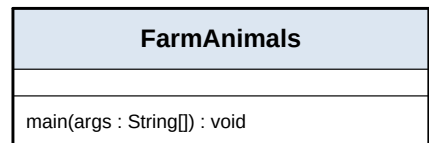
Step	Algorithm
1	Start the program.
2	Create a FarmAnimal class with name, stays, food, and sound fields.
3	Initialize each FarmAnimal object using the parameterized constructor.
4	Create a cow object with values Cow, Cowshed, Grass, and Moo.
5	Create a pig object with values Pig, Pigsty, Grains, and Oink.
6	Create a horse object with values Horse, Stable, Grass and Hay, and Neigh.
7	Call displayDetails() for the cow object.
8	Call displayDetails() for the pig object.
9	Call displayDetails() for the horse object.
10	Display the details of each farm animal.
11	Stop the program.

2. UML CLASS DIAGRAM

FarmAnimals creates three objects of the FarmAnimal class and calls displayDetails() for each object.



creates 3 objects



UML: FarmAnimals is the driver class. It creates cow, pig, and horse objects from FarmAnimal.

3. JAVA SOURCE CODE

File name: **FarmAnimals.java**

```
class FarmAnimal {
    String name;
    String stays;
    String food;
    String sound;

    FarmAnimal(String name, String stays, String food, String sound) {
        this.name = name;
        this.stays = stays;
        this.food = food;
        this.sound = sound;
    }

    void displayDetails() {
        System.out.println("Animal : " + name);
        System.out.println("Stays : " + stays);
        System.out.println("Eats : " + food);
        System.out.println("Sound : " + sound);
        System.out.println("-----");
    }
}

public class FarmAnimals {
    public static void main(String[] args) {

        // Creating objects
        FarmAnimal cow = new FarmAnimal(
            "Cow", "Cowshed", "Grass", "Moo"
        );

        FarmAnimal pig = new FarmAnimal(
            "Pig", "Pigsty", "Grains", "Oink"
        );

        FarmAnimal horse = new FarmAnimal(
            "Horse", "Stable", "Grass and Hay", "Neigh"
        );

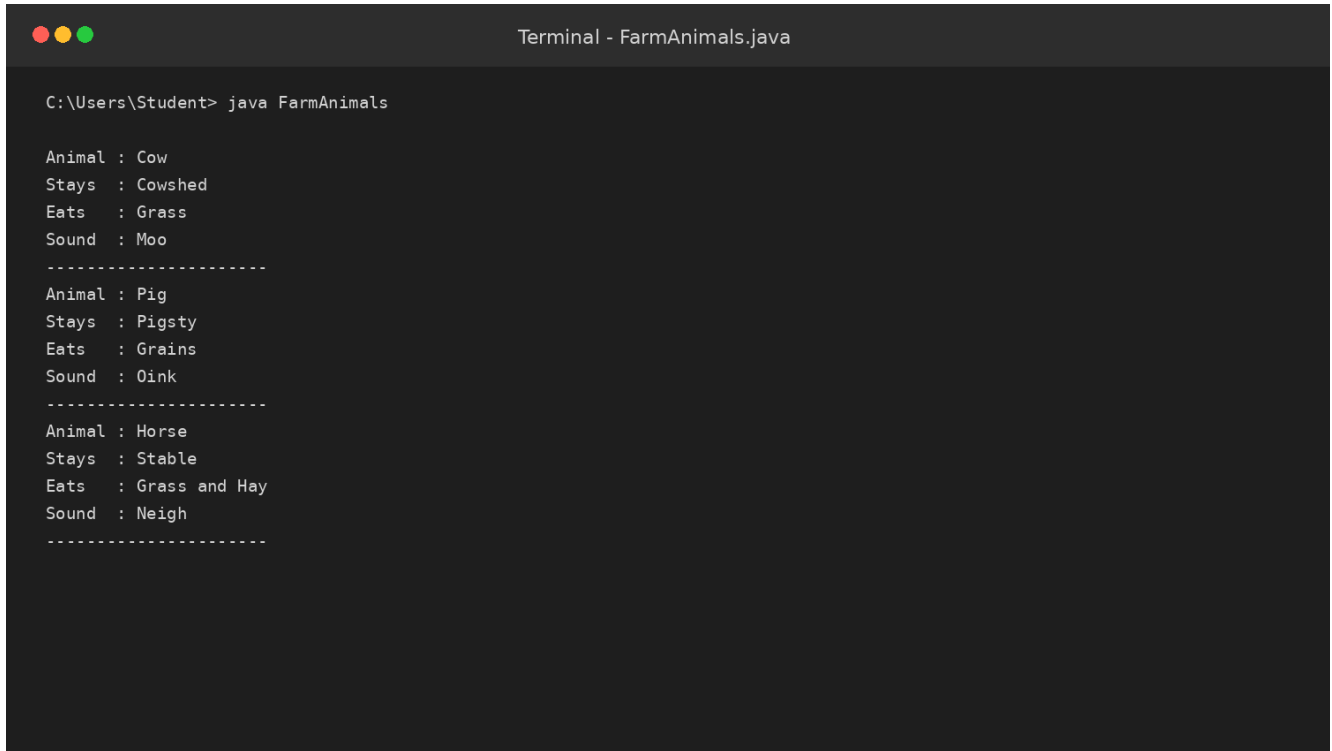
        // Displaying details
        cow.displayDetails();
        pig.displayDetails();
        horse.displayDetails();
    }
}
```

3. JAVA SOURCE CODE (CONTINUED)

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4. OUTPUT SCREEN - VS CODE STYLE

Example execution of the program:

A screenshot of a terminal window titled "Terminal - FarmAnimals.java". The terminal shows the command "C:\Users\Student> java FarmAnimals" being executed. The output displays details for three animals: Cow, Pig, and Horse, separated by dashed lines. Each animal's details include its name, where it stays, what it eats, and its sound.

```
C:\Users\Student> java FarmAnimals

Animal : Cow
Stays  : Cowshed
Eats   : Grass
Sound  : Moo
-----
Animal : Pig
Stays  : Pigsty
Eats   : Grains
Sound  : Oink
-----
Animal : Horse
Stays  : Stable
Eats   : Grass and Hay
Sound  : Neigh
-----
```

Result: Details of the cow, pig, and horse are displayed.